Lesson 1: Similar Polygons and Scale Factor

Shapes that are the same shape but not always the same size, matching their corners and sides in the right order, and using a scale factor to enlarge, reduce and find missing lengths.

1. Same Shape, Different Size

A **polygon** is a closed flat shape with straight sides: a triangle (3 sides), a quadrilateral (4 sides), a pentagon (5 sides), a hexagon (6 sides), and so on.

Two figures are **similar** if they have exactly the **same shape**. They can be different sizes. A photo and a bigger print of it are similar. A model car and the real car are similar. But if you stretch a photo in only one direction, everyone in it looks too wide or too thin. The shape has changed, so the stretched photo is **not** similar to the original.

Two polygons are **similar** when **both** of these are true:

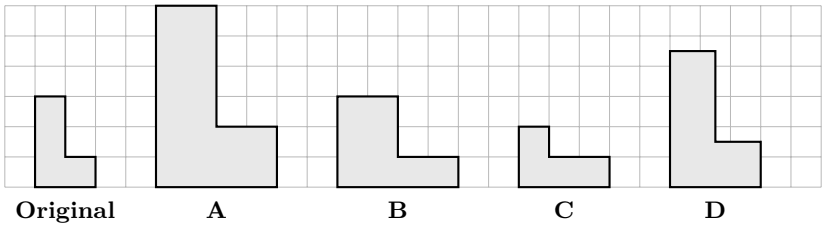
1. **Corresponding angles are equal.** The matching corners are the same size.
2. **Corresponding sides are in the same ratio.** Every side of one figure is the matching side of the other figure **multiplied by the same number**.

The symbol $\sim$ means "is similar to". Figures with the same shape **and** the same size are **congruent**. Congruent figures are similar too.

Remember. Turning or flipping a figure does not change its shape. Only stretching or squishing does.

Example 1

Each small square is 1 unit. Which copies are similar to the original?



(a) Which copies are similar to the original?

(b) For each similar copy, every length was multiplied by what number?

(c) Explain why copy B is **not** similar to the original.

2. Corresponding Parts and Naming Order

Corresponding parts are parts in matching positions: the matching angles and the matching sides. When we write $ABCD \sim PQRS$, the **order of the letters** tells you what matches. The 1st letter goes with the 1st letter, the 2nd with the 2nd, and so on.

$$\begin{array}{cccc} A & B & C & D \\ \updownarrow & \updownarrow & \updownarrow & \updownarrow \\ P & Q & R & S \end{array}$$

Angles: $\angle A$ and $\angle P$, $\angle B$ and $\angle Q$, $\angle C$ and $\angle R$, $\angle D$ and $\angle S$

Sides: AB and PQ, BC and QR, CD and RS, DA and SP

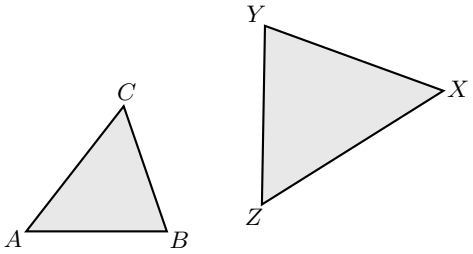
If $ABCD \sim PQRS$, then $\angle A = \angle P,\ \angle B = \angle Q,\ \angle C = \angle R,\ \angle D = \angle S$ and $\dfrac{PQ}{AB} = \dfrac{QR}{BC} = \dfrac{RS}{CD} = \dfrac{SP}{DA}$.

Watch out. The second figure is often turned or flipped. Do not match parts by where they sit on the page. Write the two names one above the other and read down the columns.

Quick refresher. From Lesson 5.3: the angles in a triangle add up to 180°, in a quadrilateral 360°, and in a pentagon 540°.

Example 2

$\triangle ABC \sim \triangle XYZ$. The second triangle is turned around.



(a) Complete the table of corresponding parts.

In $\triangle ABC$	$\angle A$	$\angle B$	$\angle C$
In $\triangle XYZ$			

In $\triangle ABC$	side AB	side BC	side CA
In $\triangle XYZ$			

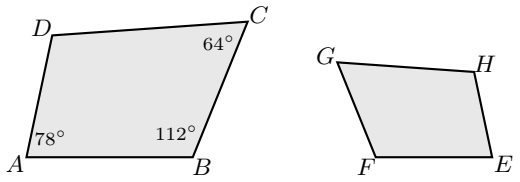
(b) $\angle A = 52°$ and $\angle B = 71°$. Find $\angle C$, $\angle X$, $\angle Y$ and $\angle Z$.

(c) Write the three equal ratios of corresponding sides (new over original).

Example 3

Quadrilateral $ABCD \sim$ quadrilateral $EFGH$. $EFGH$ is smaller and flipped over. $\angle A = 78°$, $\angle B = 112°$ and $\angle C = 64°$.

(a) Find $\angle D$.



(b) Find $\angle E$, $\angle F$, $\angle G$ and $\angle H$.

(c) Which side of $EFGH$ corresponds to BC? _____ Which side of $ABCD$ corresponds to HE? _____

3. The Scale Factor

When a figure is enlarged or reduced to make a similar figure, every length is multiplied by the same number. This number is called the **scale factor**, k. The first figure is the **original**. The figure you make from it is the **new figure** (also called the **image**).

Scale factor: $k = \dfrac{\text{new length}}{\text{original length}}$ (use a pair of corresponding sides)

$k > 1$: an **enlargement** (the new figure is bigger) $k < 1$: a **reduction** (smaller) $k = 1$: **same size**

Watch out. Put the **new** length on top. Both lengths must be in the **same unit**, so change units first (1 m = 100 cm). Estimate: if the new figure is bigger, k must be more than 1.

Example 4

Each row gives a side of an original figure and the matching side of a similar new figure. Find the scale factor. Is it an enlargement, a reduction, or the same size?

	Original	**New**	**Scale factor k**	**Type**
(a)	4 cm	10 cm		
(b)	12 m	3 m		
(c)	15 cm	6 cm		
(d)	7 in	7 in		
(e)	2.4 ft	8.4 ft		

A scale factor can be a fraction. In (b), $k = 0.25 = \frac{1}{4}$: each new side is *one quarter* of the original side.

Example 5

(a) A school logo is 5 cm wide on a letter. On a banner in the gym, the logo is 1.2 m wide. Find the scale factor from the letter to the banner.

(b) A model of a railway boxcar is 18 cm long. The real boxcar is 15.66 m long. Find the scale factor from the real boxcar to the model.

Model scales. A model labelled **1 : 87** means model length : real length $= 1 : 87$. Every part of the model is $\frac{1}{87}$ of the real size, so $k = \frac{1}{87}$ from real to model. Other common scales: 1 : 24 and 1 : 64 for model cars.

4. Using the Scale Factor to Find Missing Sides

Once you know the scale factor, you can find any missing side of two similar figures.

Step 1. Match the corresponding sides (use the naming order).

Step 2. Find k from a pair of corresponding sides where you know **both** lengths: $k = \text{new} \div \text{original}$.

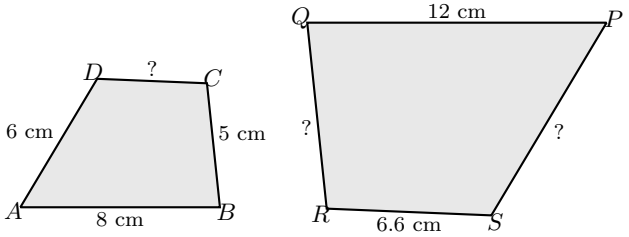
Step 3. A side of the **new** figure: **multiply**, new $= k \times$ original.
A side of the **original** figure: **divide**, original $=$ new $\div k$.

Step 4. Check: in an enlargement the new sides are longer; in a reduction they are shorter.

Another way. You can also use a proportion, as in Lesson 1.4. For a missing side x: $\dfrac{x}{5} = \dfrac{12}{8}$, so $x = 12 \times 5 \div 8 = 7.5$. Both methods give the same answer.

Example 6

$ABCD \sim PQRS$. The diagrams are not drawn to scale.



(a) Which two sides give the scale factor? Find k from $ABCD$ to $PQRS$.

(b) Find QR and SP.

(c) Find CD. Then check one answer with a proportion.

Example 7

(a) Mia has a 10 cm by 15 cm photo of her dog. She orders a similar poster print that is 60 cm on the short side.

(i) Find the scale factor.

(ii) How long is the long side of the poster?

(b) A $1:24$ model of a pickup truck is sold at a hobby shop in Red Deer.

(i) The real truck is 5.88 m long. How long is the model, in centimetres?

(ii) The model is 8.25 cm tall. How tall is the real truck, in metres?

Good to know. Areas do not grow by k. If every length is doubled ($k = 2$), the area is $2 \times 2 = 4$ times as big. The 60 cm by 90 cm poster above has $6 \times 6 = 36$ times the area of the photo.

Lesson 2: Are They Similar? Checking Angles and Side Ratios

The two tests for similar polygons (equal angles and equal side ratios), how to explain why two shapes are not similar, and why regular polygons with the same number of sides are always similar.

1. The Two Tests for Similar Polygons

Two polygons are **similar** if they have exactly the same shape. One can be bigger or smaller than the other, or turned around, but nothing is stretched or squished. Matching parts are called **corresponding** angles and **corresponding** sides. The symbol $\sim$ means "is similar to". In $ABCD \sim PQRS$ the letters are in matching order: $A \leftrightarrow P$, $B \leftrightarrow Q$, $C \leftrightarrow R$, $D \leftrightarrow S$, so side $AB \leftrightarrow PQ$, and so on.

Two polygons are similar only if BOTH tests pass.

Test 1 (angles): every pair of corresponding angles is **equal**.

Test 2 (sides): every pair of corresponding sides has the **same ratio**. Divide (side of the new shape) $\div$ (matching side of the original) for every pair. If you get the same number every time, that number is the **scale factor** k.

If even **one** pair fails, the polygons are **not similar**.

Step 1: Match the corresponding vertices (use the naming order, or match the angles).

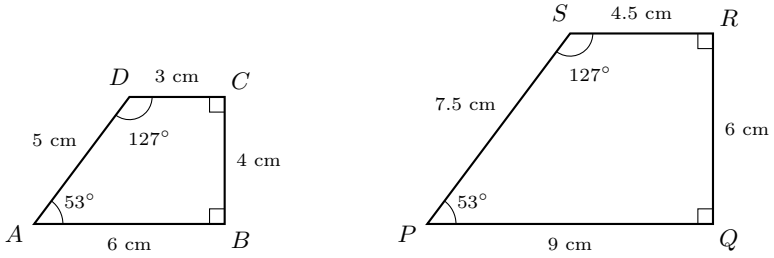
Step 2: Compare each pair of angles. **Step 3:** Divide each pair of sides, the same way every time.

Step 4: Decide, and write a sentence: "The polygons are (not) similar because . . . "

Watch out. Divide the same way every time: the **same shape's** side always goes on top. If you flip one ratio ($6 \div 9$ instead of $9 \div 6$) the numbers will not match, even when the shapes are similar.

Example 1

Is $ABCD \sim PQRS$? Check both tests.



Test 1: angles		
Pair	**Sizes**	**Equal?**
$\angle A$ and $\angle P$		
$\angle B$ and $\angle Q$		
$\angle C$ and $\angle R$		
$\angle D$ and $\angle S$		

Test 2: sides		
Pair	$PQRS$ **side** $\div$ $ABCD$ **side**	**Ratio**
PQ and AB		
QR and BC		
RS and CD		
SP and DA		

Similar? _____ Scale factor $k =$ _____ Enlargement or reduction? _____

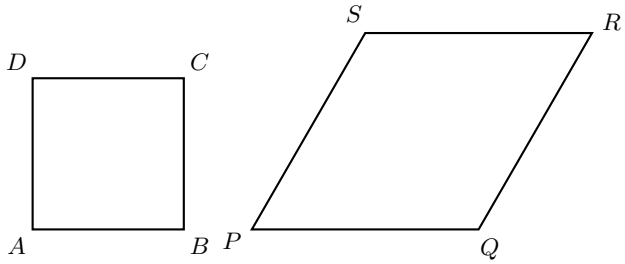
2. Checking the Angles

Making a shape bigger or smaller never changes its angles. So in similar polygons, corresponding angles are **equal**. You can measure the angles with a protractor, read them from the diagram, or work out a missing one.

Quick refresher. The angles inside a polygon with n sides add up to $(n-2) \times 180°$ (Unit 5). A quadrilateral (4 sides): 360°. A pentagon (5 sides): 540°. A hexagon (6 sides): 720°.

Example 2

Square $ABCD$ and rhombus $PQRS$ are drawn full size. (A **rhombus** has four equal sides.) Print this booklet at 100 % (actual size, not "fit to page") so your measurements match the answers.



(a) Measure one side of each shape to the nearest tenth of a centimetre.

$AB =$ _____ $PQ =$ _____

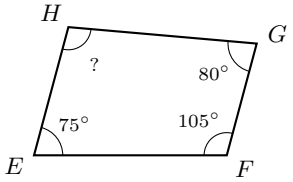
(b) All the sides of each shape are equal. Find the ratio for every pair of sides.

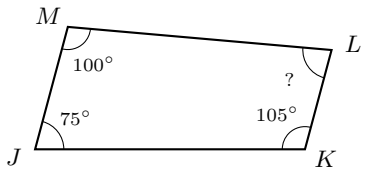
(c) Measure the angles. $\angle A =$ _____ $\angle P =$ _____
$\angle Q =$ _____ $\angle R =$ _____ $\angle S =$ _____

(d) Are the square and the rhombus similar? Explain.

Example 3

Quadrilaterals $EFGH$ and $JKLM$ have the angles shown. Their side lengths are not given.





(a) Find $\angle H$ and $\angle L$.

(b) Do all the corresponding angles match?

(c) Can you say $EFGH \sim JKLM$? Explain.

(d) Two pentagons have three pairs of equal angles: 100°, 120° and 95°. The other two angles of each pentagon are not given. All the side ratios are 2. Is that enough to say the pentagons are similar? Explain.

3. Checking the Side Ratios

Equal angles are not enough. The sides must all grow (or shrink) by the **same scale factor**. Divide each pair of corresponding sides. If all the ratios are the same, Test 2 passes.

Rectangles: every rectangle has four 90° angles, so Test 1 always passes. Only the sides decide.
Method 1 (between): long ÷ long and short ÷ short. Similar if the two answers are equal (that is k).
Method 2 (within): long side ÷ short side for **each** rectangle. Similar if the two answers are equal.

Watch out. Match **short with short** and **long with long**, even if one rectangle is turned on its side. If the units are different (in and cm, or cm and m), convert first for Method 1. Method 2 works without converting, because both sides of one rectangle have the same units.

Calculator. If a ratio does not come out even, round it to 2 decimal places. $10 \div 7$ [ENTER] gives $1.428571\ldots \approx 1.43$. If the rounded ratios are different, the shapes are not similar.

Example 4

Kiara has a 5 in by 7 in photo. She wants a bigger copy that has **exactly** the same shape (nothing cut off). Which of these are similar to her photo? Use Method 2 (long ÷ short). Her photo: $7 \div 5 = 1.4$.

Print size	Long ÷ short	Similar?
(a) 10 in by 14 in		
(b) 8 in by 10 in		
(c) 0.5 m by 0.7 m (a poster)		

(d) Check (a) with Method 1. What is the scale factor?

(e) What happens if Kiara orders the 8 in by 10 in print?

Example 5

The Canadian flag is twice as long as it is wide. A school has four flags. Which flag is **not** similar to the others? Show your work.

Flag	Long ÷ short
(a) 90 cm by 180 cm	
(b) 1.5 m by 3 m	
(c) 60 cm by 100 cm	
(d) 2 ft by 4 ft	

The same side test works for **3-D objects** such as boxes and cans: divide every pair of matching measurements (length, width, height, or radius and height).

Example 6

(a) Two shipping boxes measure 20 cm by 30 cm by 40 cm and 30 cm by 45 cm by 60 cm. Are they similar?

(b) One soup can has a radius of 4 cm and a height of 10 cm. A bigger can has a radius of 6 cm and a height of 12 cm. Are the cans similar?

4. Explaining Why Two Polygons Are Not Similar

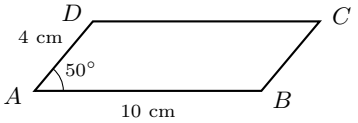
To show two polygons are **not** similar, you only need **one** pair of angles that are not equal, or **one** side ratio that does not match. Name it in your answer.

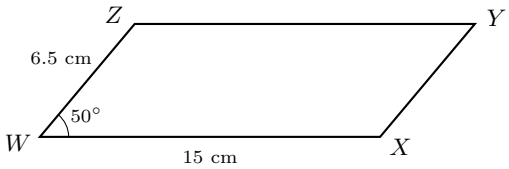
Three common traps		
A square and a rectangle (not a square)	angles all match (90°)	but the side ratios do **not**
A square and a rhombus (not a square)	side ratios all match	but the angles do **not**
Adding the same amount to every side	(a frame, a border)	the side ratios change
Sentence frame: The polygons are not similar because _____ is not equal to _____.		

Quick refresher. In a **parallelogram**, opposite sides are parallel and equal. Opposite angles are equal, and two angles next to each other add to 180° (interior angles on the same side of a transversal, Unit 5).

Example 7

Are parallelograms $ABCD$ and $WXYZ$ similar? (Not drawn to scale.)





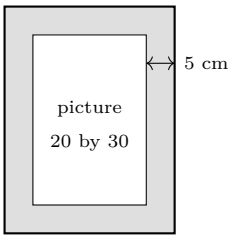
(a) Find all four angles of each parallelogram.

(b) Find the side ratios.

(c) Use the sentence frame to answer the question.

Example 8

(a) A 20 cm by 30 cm picture is put on a mat with a 5 cm border on every side. Find the outside size of the mat. Is the matted picture similar to the picture alone? Show your work.



(b) Owen checks if a 24 cm by 40 cm rectangle and a 50 cm by 30 cm rectangle are similar. Find his mistake, and fix it.

Owen's work: $50 \div 24 \approx 2.08$ and $30 \div 40 = 0.75$. They are not equal, so they are not similar.

5. Regular Polygons

A **regular polygon** has all its sides equal and all its angles equal: an equilateral triangle, a square, a regular pentagon, a regular hexagon (6 sides), a regular octagon (8 sides, like a stop sign), and so on.

Take two regular polygons with the **same number of sides**. Their angles are all the same size: each angle is $(n-2) \times 180^\circ \div n$. And since all the sides of each shape are equal, every side ratio is the same number. Both tests pass every time!

Two regular polygons with the same number of sides are always similar.

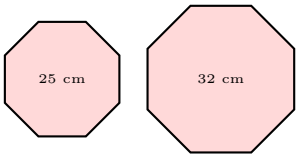
Always similar: two squares, two equilateral triangles, two regular hexagons, ...

Sometimes similar (you must check): two rectangles, two rhombuses, two parallelograms, two pentagons that are not regular.

Never similar: two polygons with a different number of sides (a square and a pentagon).

Example 9

(a) A stop sign is a regular octagon with sides of 25 cm. A larger stop sign for a highway has sides of 32 cm. Are the signs similar? Find the size of each angle and the scale factor.



(b) A tile is a regular hexagon with sides of 10 cm. Another tile is a regular pentagon with sides of 10 cm. Are they similar? Explain.

(c) Write *always*, *sometimes* or *never*.

Two squares are similar. _____

Two rectangles are similar. _____

A square and an equilateral triangle are similar. _____

Two rhombuses are similar. _____

Lesson 3: Drawing Enlargements and Reductions

Drawing a polygon that is similar to a given one: on grid paper, with a ruler and protractor, and by projecting from a point.

1. Multiply Every Length, Keep Every Angle

An **enlargement** is a bigger copy of a figure. A **reduction** is a smaller copy. For the copy to be **similar** (the same shape), every length must be multiplied by the *same* number, the **scale factor** k, and every angle must stay the same.

Drawing a similar figure

New length = $k\times$ original length Original length = new length $\div k$

Every angle stays the same size.

$k > 1$: enlargement (bigger) $k < 1$: reduction (smaller) $k = 1$: same size

Watch out. Adding the same amount to every side does **not** make a similar figure. A 2 cm by 4 cm rectangle with 2 cm added to each side becomes 4 cm by 6 cm. The short side doubled ($\times 2$) but the long side did not ($\times 1.5$), so the shape changed. Always **multiply**.

Example 1

(a) Quadrilateral $ABCD$ has $AB = 6$ cm, $BC = 10$ cm, $CD = 8$ cm, $DA = 4$ cm and $\angle A = 75°$. Complete the table for each copy.

k	**Enlargement or reduction?**	AB	BC	CD	DA	$\angle A$
3						
0.5						
1.25						

(b) A copy made with $k = 2.5$ has a side of 20 cm. How long was that side on the original?

Example 2

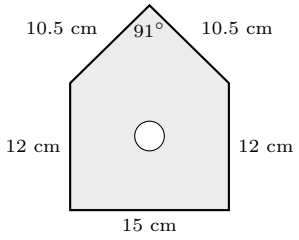
Ren is building a bigger copy of a birdhouse. The front of the old birdhouse is shown (not to scale). The new front will have a base of 24 cm.

(a) Find the scale factor.

(b) Find the lengths of the new walls and roof edges.

(c) What is the angle at the peak of the new roof?

(d) The old birdhouse is 14 cm deep (front to back). How deep should the new one be?



2. The Grid Method

Grid paper makes enlargements easy, because you can **count squares** instead of measuring. Walk around the shape from corner to corner and write each move: how many squares **right or left**, and how many **up or down**. Then multiply every count by k.

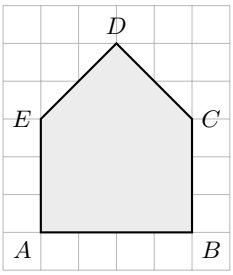
Grid method

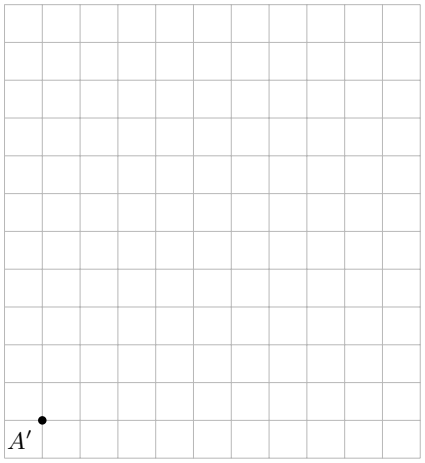
Step 1. Pick a starting corner on the original. Mark a starting dot on the new grid.
Step 2. Count the move to the next corner (right/left and up/down).
Step 3. Multiply each count by k. Make that move on the new grid and mark the corner.
Step 4. Repeat all the way around. You should land back on the starting dot.
Step 5. Join the corners with a ruler.

Watch out. Don't count squares *along* a slanted side. A slanted side that goes 2 right and 1 up becomes 4 right and 2 up when $k = 2$. Multiply **both** counts. If $k = 1.5$ and a count is odd, the corner lands in the middle of a square. That's fine: $3 \times 1.5 = 4.5$ squares.

Example 3

Draw an enlargement of the house shape $ABCDE$ with a scale factor of 2. Start at the dot A' (say "A prime"; it matches A). Fill in the table as you go.

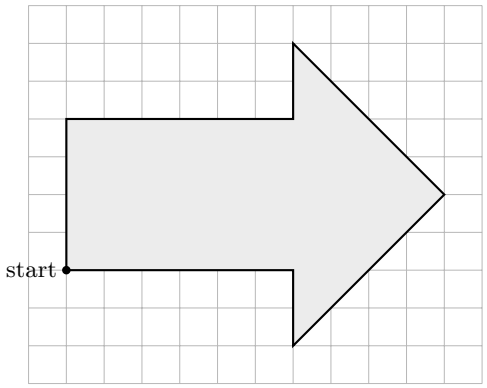


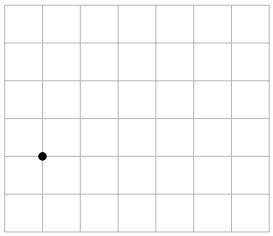


Side	**Original move**	**New move** ($\times 2$)
$A \to B$	4 right	
$B \to C$		
$C \to D$		
$D \to E$		
$E \to A$		

Example 4

A road crew's detour sign is an arrow. Draw a reduction of the arrow with a scale factor of $\frac{1}{2}$ (that is, $k = 0.5$). Start at the dot.

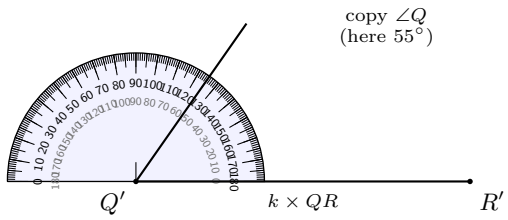




3. The Ruler and Protractor Method

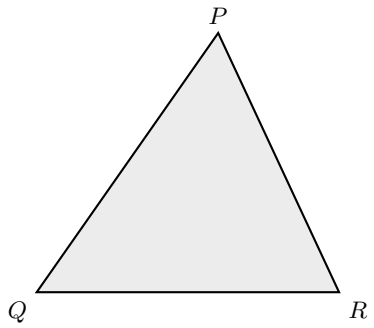
When a figure is not on a grid, **measure** it. Multiply each side length by k, and copy each angle with a protractor (Unit 5). A triangle only needs one side and the two angles at its ends: the other two sides are drawn for you when the arms meet.

Ruler and protractor method (a triangle)
Step 1. Measure one side and the two angles at its ends.
Step 2. Multiply the side by k. Draw the new side with a ruler.
Step 3. At each end of the new side, use the protractor to draw an arm at the same angle as the original.
Step 4. Extend the two arms until they meet. That point is the third corner.
Step 5. Check: measure the third angle (it should match) and the other sides (each should be k times the original).
More sides? Draw a side, copy the angle at its end, measure the next side ($\times k$) along the new arm, and keep going.



Example 5

(a) Measure side QR, $\angle Q$ and $\angle R$ of $\triangle PQR$. (Print this booklet at 100 % (actual size, not "fit to page") so your measurements match the answers.)

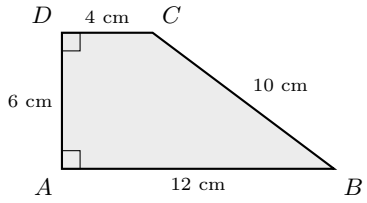


(b) In the space, draw $\triangle P'Q'R'$, an enlargement of $\triangle PQR$ with $k = 1.5$.

(c) Check: measure $\angle P$, $\angle P'$, PQ and $P'Q'$.

Example 6

Theo sketched the side of a skateboard ramp (not to scale). Angles A and D are right angles. Draw a reduction of $ABCD$ with $k = 0.5$. Then measure $B'C'$ to check your drawing.



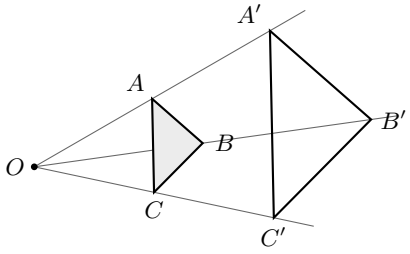
4. The Projection Method

The **projection method** works like a projector or a flashlight making a shadow on a wall: everything spreads out from one point. Pick a point O, the **centre**. Draw a ray from O through each corner. Each new corner goes on the same ray, k times as far from O.

Projection method

Step 1. Draw a ray from O through each corner (A, B, C, ...).
Step 2. Measure OA. Multiply by k. Mark A' on the same ray, that far from O: $OA' = k \times OA$.
Step 3. Do the same for every corner.
Step 4. Join the new corners in the same order.

If $k > 1$, the new corners are *past* the old ones. If $k < 1$, they are *between* O and the old ones. O can be outside or inside the shape.



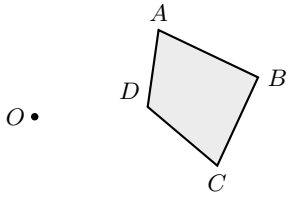
$k = 2$: $OA' = 2 \times OA$

Watch out. Measure from O, not from the old corner. With $k = 3$ and $OA = 2$ cm, A' is 6 cm from O (only 4 cm past A).

Example 7

Draw an enlargement of $ABCD$ with centre O and $k = 2$. Fill in the table as you go.

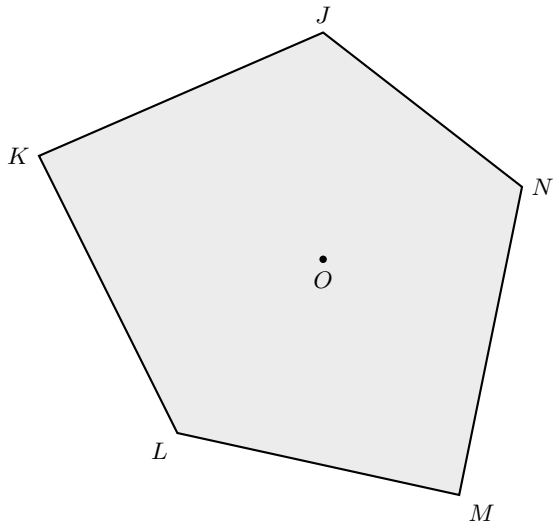
Corner	**From O**	**New: $\times 2$**
A	$OA =$	$OA' =$
B	$OB =$	$OB' =$
C	$OC =$	$OC' =$
D	$OD =$	$OD' =$



Example 8

Draw a reduction of the pentagon $JKLMN$ with centre O (inside the shape) and $k = \frac{1}{2}$.

Corner	**From O**	**New: $\times\frac{1}{2}$**
J		
K		
L		
M		
N		



5. Photos, Posters and Checking Your Drawing

A photo is a rectangle. A print or poster is similar to the photo only if **both** sides are multiplied by the same k. Common photo print sizes (in inches) are 4×6, 5×7, 8×10 and 11×14, and they are *not* all similar. If a print is not similar to the photo, part of the picture gets **cropped** (cut off), or the picture gets stretched.

Fitting an enlargement on paper or in a frame

Find k for each direction: $k =$ space available $\div$ photo length.
Use the **smaller** k. Then the whole picture fits, with some space left over on one side.

Checking a drawing

1. Measure each pair of matching angles. They should be equal.
2. Divide each new side by its matching old side. Every answer should be k.
Measuring isn't perfect: being off by about 1 mm or 1° is fine.

Example 9

(a) Mei has a 4 in by 6 in photo. She orders an enlargement with $k = 3$. What size is the print?

(b) Is an 8 in by 10 in print similar to Mei's 4 in by 6 in photo? Explain.

(c) The school's art club has a 15 cm by 10 cm photo. They want the biggest similar enlargement that fits on a 60 cm by 45 cm sheet of poster paper. Find k, the size of the poster, and how much paper is left over.

Example 10

Jada was asked to enlarge quadrilateral $WXYZ$ with $k = 3$. Here are her measurements.

Side	WX	XY	YZ	ZW
Original	2.0 cm	3.0 cm	2.5 cm	4.0 cm
Jada's copy	6.0 cm	9.0 cm	8.5 cm	12.0 cm
Copy ÷ original				

(a) Is Jada's copy similar to the original? Which side is wrong, and how long should it be?

(b) The sides are fixed. What else must Jada check?

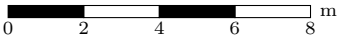
Lesson 4: Scale Diagrams: Maps and Floor Plans

Reading the scale on a map, plan or model, using proportions to change diagram lengths into real lengths and back, and choosing a scale to draw a floor plan.

1. What Is a Scale Diagram?

A **scale diagram** is an accurate drawing of a real object in which every length is changed by the same amount. Maps, floor plans, blueprints and model cars are smaller than the real thing: they are **reductions**. A drawing of an insect or a tiny machine part can be bigger than the real thing: an **enlargement**. A scale diagram is **similar** to the real object: it has the same shape and the same angles.

The **scale** tells you how a length on the diagram compares with the real length. A scale can be written three ways. All three below say the same thing.

In words	**As a ratio**	**As a bar scale**
1 cm represents 2 m	1 : 200	

Scale = diagram length : real length (the diagram comes first)

A **ratio scale** has no units because both numbers are in the **same unit**. 1 : 200 means 1 cm on the plan is 200 cm in real life (or 1 mm is 200 mm, or 1 in is 200 in).

Units: 1 cm = 10 mm 1 m = 100 cm 1 km = 1000 m = 100 000 cm 1 ft = 12 in

Words to ratio: change the real length to the same unit as the diagram length, then write diagram : real. **Ratio to words:** $1 : n$ means 1 cm represents n cm. Change n cm to metres or kilometres so it is easy to picture.

Watch out. 1 : 200 and 200 : 1 are very different. 1 : 200 is a reduction (a plan: the real room is bigger). 200 : 1 is an enlargement (the drawing is bigger). In Lesson 6.1 you met the scale factor: a scale of 1 : 200 means $k = \frac{1}{200} = 0.005$.

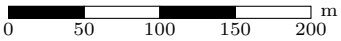
Example 1

Complete the table.

	Scale in words	As a ratio	Reduction or enlargement?
(a)	1 cm represents 5 m		
(b)	1 cm represents 3 km		
(c)	1 in represents 8 ft		
(d)	2 cm represents 1 mm		
(e)		1 : 250	
(f)		1 : 40 000	

Example 2

A park map has this bar scale. Use a ruler.



(a) How long is one block of the bar? Write the scale in words.

(b) Write the scale as a ratio.

(c) Ana zooms in on the map on her phone, so everything is twice as long, including the bar. Is the bar scale still correct? Would a printed ratio "1 : 5000" still be correct? Explain.

2. Finding Real Lengths from a Diagram

You can find a real length two ways. Both use what you learned about proportions in Unit 1.

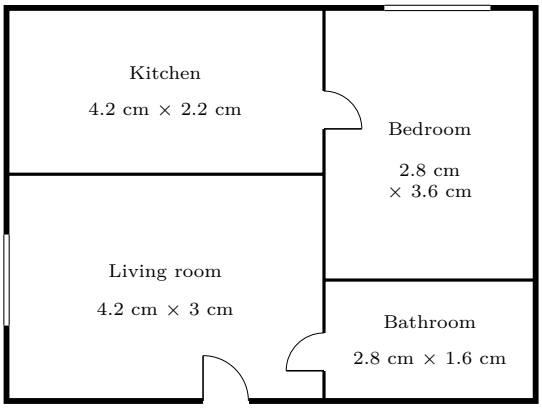
Method 1: proportion. $\frac{\text{diagram}}{\text{real}} = \frac{\text{diagram}}{\text{real}}$ Put the scale on one side and your length on the other. Units in the same position must match.

Method 2: multiply. real length = diagram length × (real length for 1 unit of the diagram)
If 1 cm represents 1.5 m, then 4 cm represents $4 \times 1.5 = 6$ m. With a ratio $1 : n$, multiply by n, then change the units.

Map shortcut. To change centimetres to kilometres, divide by 100 000 (move the decimal point 5 places left). So 1 : 2 000 000 means 1 cm represents 20 km, and 1 : 50 000 means 1 cm represents 0.5 km.

Example 3

This plan of an apartment is drawn to scale: 1 cm represents 1.5 m. The plan lengths are shown.



(a) Use a proportion to find the real length and width of the living room.

(b) Use the multiply method to complete the table.

Room	**Plan (cm)**	**Work**	**Real (m)**
Kitchen	4.2×2.2		
Bedroom	2.8×3.6		
Bathroom	2.8×1.6		
Whole apartment	7×5.2		

(c) New flooring is going into the living room. Find the real area of its floor.

Watch out. For a real area, change the *lengths* to real lengths first, then multiply. The plan area is $4.2 \times 3 = 12.6$ cm^2, but $12.6 \times 1.5 = 18.9$ m^2 is **wrong**. Areas do not grow by the scale. (Did you know? Here they grow by $1.5 \times 1.5 = 2.25$.)

Example 4

Find each real length. Say whether you multiplied or divided, and why.

(a) An Alberta road map has a scale of $1 : 2\,000\,000$. Lethbridge and Medicine Hat are 7.9 cm apart on the map. How far apart are they really (in a straight line)?

(b) A model car is built at a scale of $1 : 64$. The model is 7.5 cm long. How long is the real car, in metres?

(c) A science book shows an ant at a scale of $8 : 1$. The ant in the picture is 6 cm long. How long is the real ant, in millimetres?

3. Finding Diagram Lengths from Real Lengths

Now go the other way: you know the real length and need the length to draw.

Diagram length = real length ÷ (real length for 1 unit of the diagram), or use a proportion.
With a ratio $1 : n$: change the real length to the diagram's unit (usually cm), then divide by n.
Check: for a reduction, the diagram length must be *smaller* than the real length.

Example 5

Carpenters' blueprints often use the scale $\frac{1}{4}$ in represents 1 ft. Liam is drawing the blueprint for a garage that is 24 ft long and 22 ft wide, with a garage door 9 ft wide.

(a) How many feet does 1 in on the blueprint represent?

(b) Find the length, width and door width on the blueprint.

Example 6

A hobby shop sells models at different scales. Find each model length to the nearest tenth of a centimetre.

(a) A pickup truck is 5.9 m long. How long is a $1 : 64$ model of it?

(b) Model trains in "HO scale" are $1 : 87$. A real grain boxcar is 15.2 m long. How long is the HO model?

4. Choosing a Scale and Drawing a Floor Plan

Before you draw, choose a scale that makes the drawing fit the paper but is not tiny. Pick an easy number so the dividing is simple.

Choosing a scale
Step 1. Find the longest real length (and the longest real width).
Step 2. Measure the space you have on the paper.
Step 3. Divide: real length ÷ space. Do it for the length and the width, and use the **bigger** answer.
Step 4. Round **up** to an easy number of real units for each 1 cm (such as 0.5, 1, 2, 2.5, 5 or 10). Rounding down would make the drawing too big.
Step 5. Divide every real length by the scale, draw with a ruler, and write the scale on the drawing.

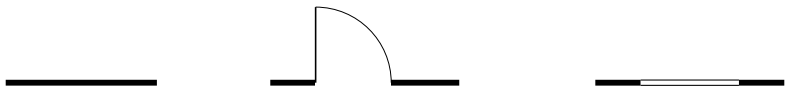
Example 7

Coach Diaz wants to draw a hockey rink, 61 m long and 26 m wide, in a space on a whiteboard handout that is 15 cm long and 10 cm wide.

(a) Choose an easy scale so the rink fits.

(b) Find the length and width of the rink on the drawing. Why would "1 cm represents 4 m" not work?

A **floor plan** is a scale diagram of a building seen from above. These symbols are used on most plans:

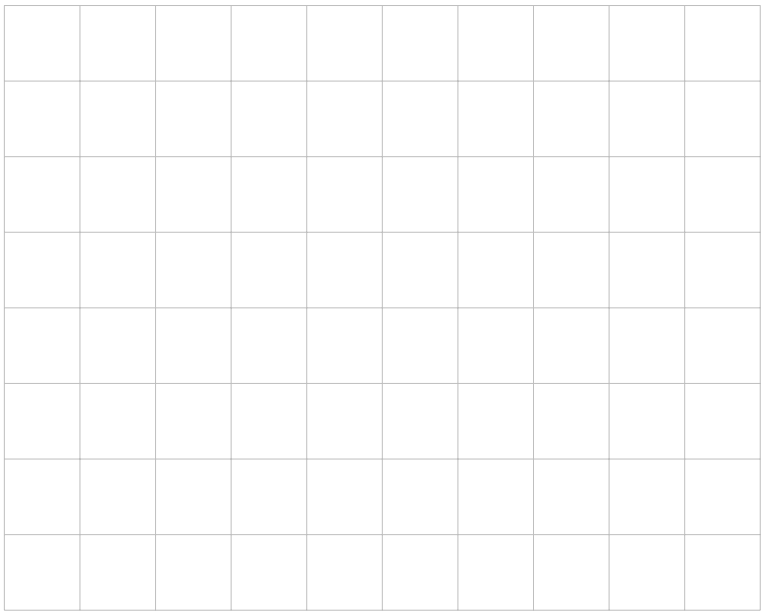


wall door (and its swing) window

Example 8

Draw a floor plan of Maya's bedroom using the scale 1 cm represents 0.5 m. The room is 4.5 m long and 3.6 m wide. It has a 0.9 m door near one corner of a long wall, a 1.5 m window in the middle of the other long wall, a bed 2.0 m by 1.5 m, and a dresser 1.2 m by 0.5 m.

Item	**Real (m)**	**Plan (cm)**
Room length	4.5	
Room width	3.6	
Door	0.9	
Window	1.5	
Bed	2.0×1.5	
Dresser	1.2×0.5	



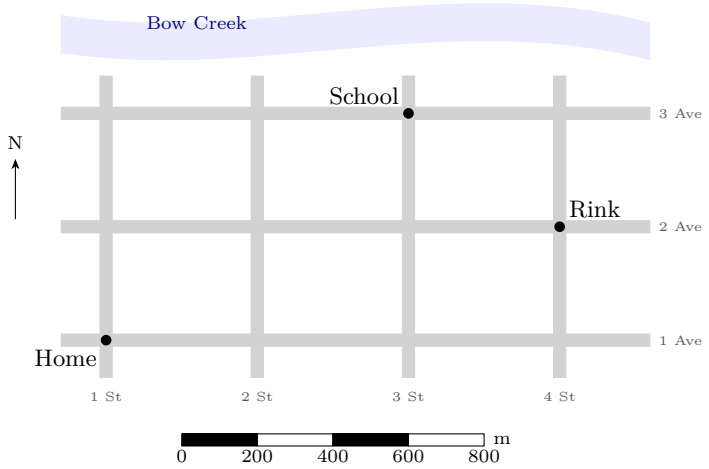
5. Maps: Distances Along a Route

The **straight-line distance** between two places is measured with a ruler from one point to the other. People usually travel along roads or trails, so the **route distance** is longer. For a route, measure each straight piece, **add** the pieces, then change the total to a real distance once. For a curved trail, lay a piece of string along it, then measure the string.

Measuring. The maps and plans in this booklet are drawn to scale. Measure them on the printed page with a ruler, to the nearest millimetre (0.1 cm). Print this booklet at 100 % (actual size, not "fit to page") so your measurements match the answers.

Example 9

This map of a small town is drawn to scale. Streets run north–south and avenues run east–west. Use a ruler.



(a) Write the scale in words and as a ratio.

(b) Jada walks from home to school along the streets: east on 1 Ave, then north on 3 St. Find her real walking distance.

(c) Find the straight-line distance from home to school.

(d) After school she walks to the rink: south on 3 St, then east on 2 Ave. How far is that?

(e) Jada walks at about 5 km/h. About how many minutes does her walk from home to school take?

Lesson 5: Similar Triangles

Two pairs of equal angles are enough to show that two triangles are similar; then a proportion finds any missing side.

1. The AA Test for Similar Triangles

Remember: two shapes are **similar** if they have the same shape. One is an enlargement or a reduction of the other. The parts that match are called **corresponding** parts: corresponding angles are equal, and corresponding sides are in the same ratio.

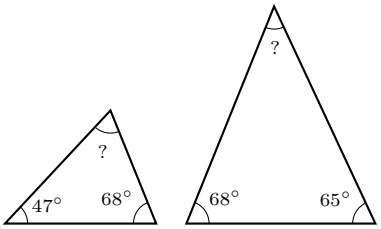
For most polygons you must check the angles *and* the sides. Triangles are easier. The three angles of any triangle add to 180°. So if two angles of one triangle match two angles of another triangle, the third angles must match too. And when all the angles match, the sides are always in the same ratio.

AA test (angle–angle): if two angles of one triangle are equal to two angles of another triangle, the triangles are **similar**.
To find the third angle of a triangle: third angle $= 180° -$ (first angle + second angle).

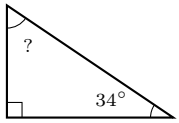
Watch out. The equal angles do not have to be in the same place on the page. A triangle can be turned or flipped over and still be similar. Match the angles by their *size*, not by where they are.

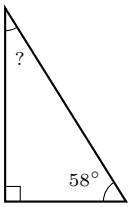
Example 1

Find the missing angles in each triangle. Then decide: are the two triangles similar? (In part (c), both triangles are **isosceles**: two equal sides, so the two angles at the base are equal.)

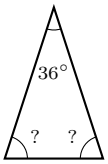


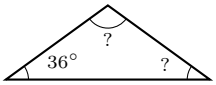
(a)





(b)

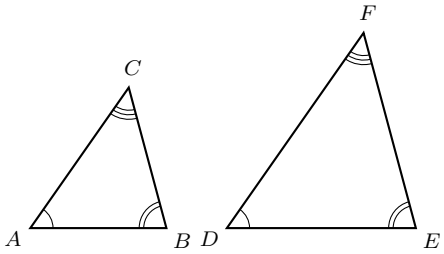




(c)

2. Naming Similar Triangles in Order

We write $\triangle ABC \sim \triangle DEF$ and say "triangle ABC is similar to triangle DEF." The **order of the letters** tells you which parts match: the 1st letters match, the 2nd letters match and the 3rd letters match.

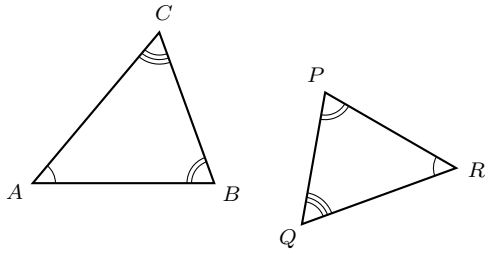


If $\triangle ABC \sim \triangle DEF$, then
equal angles: $\angle A = \angle D,\ \angle B = \angle E,\ \angle C = \angle F$
equal ratios: $\dfrac{DE}{AB} = \dfrac{EF}{BC} = \dfrac{DF}{AC} = k$, the **scale factor**.

Matching angles are often marked with the same number of arcs. A side and the angle **opposite** it (across the triangle) go together: if $\angle A$ matches $\angle D$, then the side across from A (that is BC) matches the side across from D (that is EF).

Example 2

The two triangles are similar. Matching angles have the same number of arcs.



(a) Complete: $\triangle ABC \sim \triangle$_____

(b) $\angle A = \angle$____ $\angle B = \angle$____ $\angle C = \angle$____

(c) Complete the equal ratios: $\frac{RP}{AB} = \frac{\overline{\quad}}{BC} = \frac{\overline{\quad}}{AC}$

(d) Side AB is across from $\angle C$. The matching side is across from $\angle$____, so it is side _____.

3. Finding a Missing Side

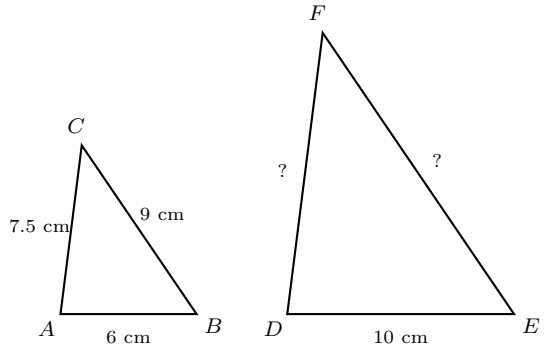
Once you know two triangles are similar, you can find a missing side with the scale factor or with a proportion (Unit 1).

Step 1. Match the vertices (write $\triangle$_____ $\sim \triangle$_____).
Step 2. Find a pair of matching sides where you know *both* lengths.
Step 3. Use the scale factor $k = \dfrac{\text{new side}}{\text{matching original side}}$ (multiply by k to go to the new triangle, divide to come back), **or** write a proportion with the missing side on top and solve.
Step 4. Check: the bigger triangle must have the bigger sides.

Calculator. Do not round k. For a side of 9 cm and $k = \frac{10}{6}$, key in [9] [×] [10] [÷] [6] [ENTER]. Round only the final answer.

Example 3

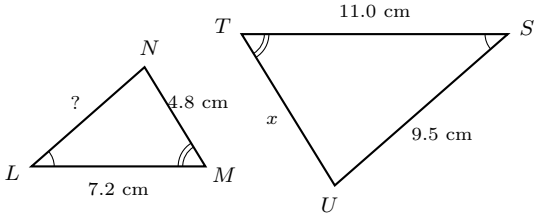
$\triangle ABC \sim \triangle DEF$. Find the lengths of EF and DF.



Example 4

$\triangle LMN \sim \triangle STU$. The second triangle is turned. Find each length to the nearest tenth of a centimetre.

(a) Find x (side TU). (b) Find side LN.



Watch out. Keep the triangles in the same places in a proportion: the same triangle on top on both sides. $\frac{x}{4.8} = \frac{11.0}{7.2}$ is right (big over small on both sides). $\frac{x}{4.8} = \frac{7.2}{11.0}$ mixes them up and gives 3.1 cm, which is too small for the bigger triangle.

4. Right Triangles That Share an Acute Angle

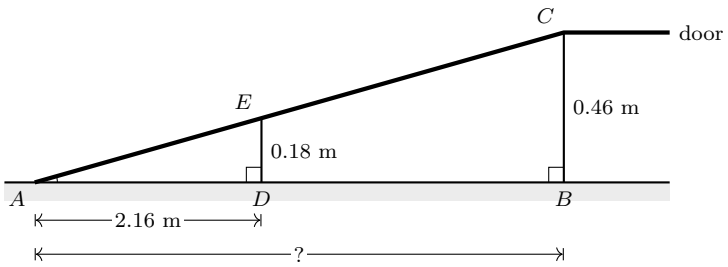
Every right triangle has a 90° angle. If two right triangles also have one **acute** angle (less than 90°) that is the same, then two angles match, so the triangles are similar by the AA test.

This happens a lot in real life. A ramp, a roof or a slide makes a long, thin right triangle with the ground. A vertical post under it cuts off a smaller right triangle. The two triangles **share** the acute angle where the slope meets the ground.

Two right triangles that share an acute angle (or have equal acute angles) are **similar**: the right angles match, the shared angles match, so by AA the triangles are similar.

Example 5

A wheelchair ramp starts at point A on level ground and rises steadily to a door. A vertical support post DE is 2.16 m from A along the ground, and it is 0.18 m tall. (The diagram is not to scale.)



(a) Complete: $\triangle ADE$ and $\triangle ABC$ are similar because both have a _____ angle, and they share $\angle$_____.

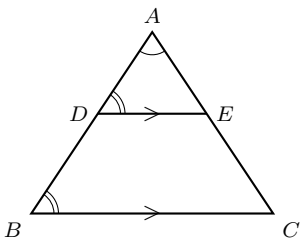
(b) The door is 0.46 m above the ground. How far from A, along the ground, must the ramp end (AB)? Round to the nearest hundredth of a metre.

(c) A second post will stand 4.00 m from A. How tall must it be, to the nearest hundredth of a metre?

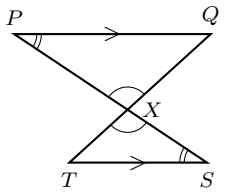
5. Nested Triangles and Crossing Triangles

Nested triangles. Draw a line inside a triangle **parallel** to one side. It cuts off a small triangle at the top. The small and big triangles share the top angle, and the parallel line makes **corresponding angles** that are equal (Unit 5). So the two triangles are similar by AA.

Crossing triangles. Two line segments cross at a point and their ends are joined by two parallel lines. This makes two triangles point to point, like a bow tie. The angles where the lines cross are **vertically opposite** (equal), and the parallel lines make **alternate interior angles** (equal). So these triangles are similar by AA too.



nested: $\triangle ADE \sim \triangle ABC$



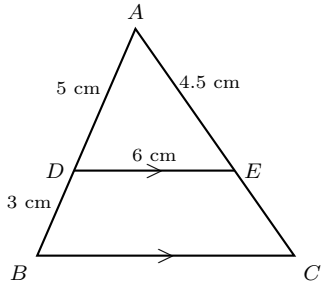
crossing: $\triangle XPQ \sim \triangle XST$

Watch out. In nested triangles, the big triangle's side is the *whole* side, not just the bottom piece. If $AD = 5$ and $DB = 3$, the big side $AB = 5 + 3 = 8$. It helps to draw the two triangles apart and label each one.

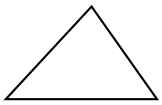
Example 6

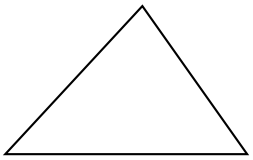
In $\triangle ABC$, $DE \parallel BC$. $AD = 5$ cm, $DB = 3$ cm, $DE = 6$ cm and $AE = 4.5$ cm.

(a) Explain why $\triangle ADE \sim \triangle ABC$.



(b) Label the two triangles, drawn apart. Put the side lengths you know on them.



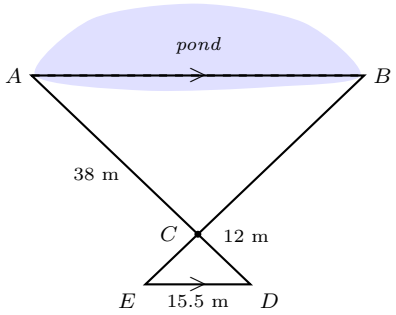


(c) Find the length of BC.

(d) Find the length of EC.

Example 7

Nadia wants to know the width AB of a pond on her farm. She puts a stake at C, so that she can walk in a straight line from A through C to D, and from B through C to E. She places D and E so that DE is parallel to AB. Then she measures: $AC = 38$ m, $CD = 12$ m and $DE = 15.5$ m.



(a) Give the reasons why $\triangle ABC \sim \triangle DEC$.

(b) Find the width of the pond, AB, to the nearest tenth of a metre.

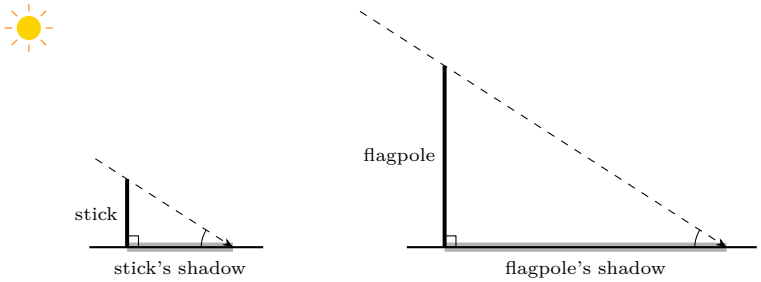
Lesson 6: Indirect Measurement: Shadows and Mirrors

Find the height of a tree, a tower or a flagpole, or the width of a river, without climbing or crossing it: measure what you can reach, then use similar triangles.

1. Indirect Measurement and the Shadow Method

Some lengths are too high or too far away to measure with a tape: a flagpole, a tree, a cell-phone tower, the width of a river. **Indirect measurement** means finding a length *without* measuring it directly. You measure lengths you can reach, find two **similar triangles** (Lesson 5), and solve a proportion.

On a sunny day, everything casts a shadow. The sun is so far away that its rays are **parallel**: at any moment, they come down at the same angle everywhere nearby.



An object standing straight up, its shadow and the sun ray make a **right triangle**. Two such triangles have

- a **right angle** at the bottom of the object (it stands straight up on level ground), and
- the **same angle** at the tip of the shadow (the sun's angle is the same for both).

Two pairs of equal angles, so the triangles are **similar** (AA). In general: **two right triangles that share the same acute angle are always similar**. So the heights and the shadows are in proportion.

The shadow method. Measure the two shadows at the **same time**, then

$$\frac{\text{height of object}}{\text{shadow of object}} = \frac{\text{height of stick}}{\text{shadow of stick}}$$

Step 1. Sketch and label both triangles. **Step 2.** Write the word proportion (heights on top, shadows on the bottom).
Step 3. Put in the numbers (same units) and cross-multiply. **Step 4.** Check that it makes sense; write a sentence.

Watch out. Shadows change all day: long in the morning, short around noon, long again in the evening. Both shadows must be measured within a few minutes of each other. The ground must be level and the objects must stand straight up.

Example 1

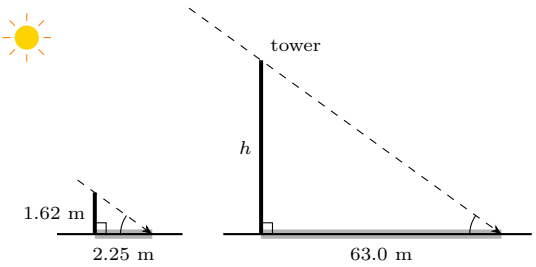
At 3:00 p.m., an upright metre stick casts a shadow 1.60 m long. At the same time, the school's flagpole casts a shadow 12.8 m long. How tall is the flagpole?

	Height	**Shadow**
Metre stick	1.00 m	1.60 m
Flagpole	h	12.8 m

Example 2

Jasmine is 1.62 m tall. Late one afternoon near Airdrie, her shadow is 2.25 m long. At the same time, a cell-phone tower casts a shadow 63.0 m long.

(a) Estimate the height of the tower.



(b) Find the height of the tower to the nearest tenth of a metre. Compare with your estimate.

(c) A light standard in the same parking lot is 9.0 m tall. How long is its shadow at that moment?

Calculator. You can do the multiply and the divide in one line. For part (b): 63 [×] 1.62 [÷] 2.25 [ENTER] gives 45.36. Round only at the end.

Example 3

On a prairie farm, a fence post sticks up 6 ft out of the level ground. Its shadow is 4 ft 6 in long. At the same time, an old wooden grain elevator casts a shadow 52 ft long.

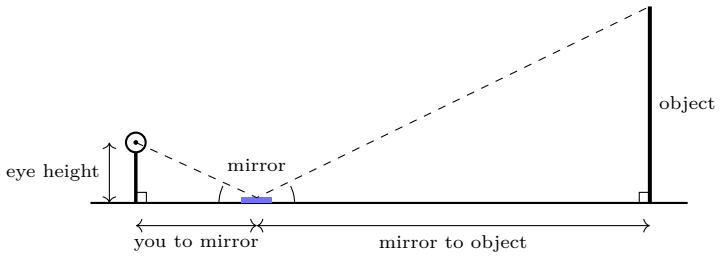
(a) Sketch the two triangles. How tall is the grain elevator, to the nearest tenth of a foot?

(b) About how tall is the elevator in metres? Use 1 ft = 0.3048 m and round to the nearest tenth.

2. The Mirror Method

On a cloudy day there are no shadows. You can use a small mirror instead. Lay the mirror flat on level ground between you and the object. Walk backwards (in a straight line away from the object) until you see the **top** of the object in the centre of the mirror. Then measure along the ground.

Light bounces off a mirror at the same angle it hits it: **angle in = angle out**. You and the object both stand straight up, so both triangles have a right angle. Two pairs of equal angles: the triangles are **similar**.



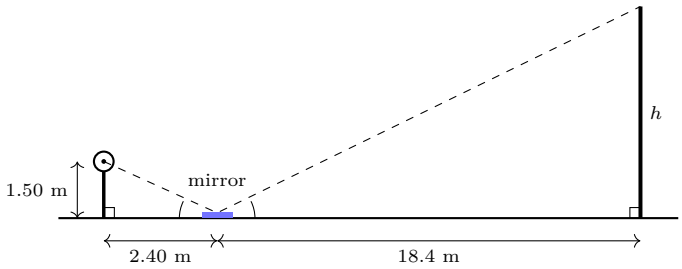
The mirror method.

$$\frac{\text{height of object}}{\text{object to mirror}} = \frac{\text{your eye height}}{\text{you to mirror}}$$

Measure to the **centre** of the mirror. Use the height of your **eyes**, not your full height: the light goes to your eyes, not the top of your head.

Example 4

Diego wants the height of the flagpole at his school. He puts a mirror on the ground 18.4 m from the base of the pole. He backs up until he sees the top of the pole in the mirror. His eyes are 1.50 m above the ground and he is 2.40 m from the mirror.



(a) How tall is the flagpole?

(b) Sara's eyes are 1.70 m above the ground. Diego leaves the mirror in the same spot. How far from the mirror must Sara stand to see the top of the pole? Round to the nearest hundredth of a metre.

3. Sighting Methods: a Ruler, or Stakes

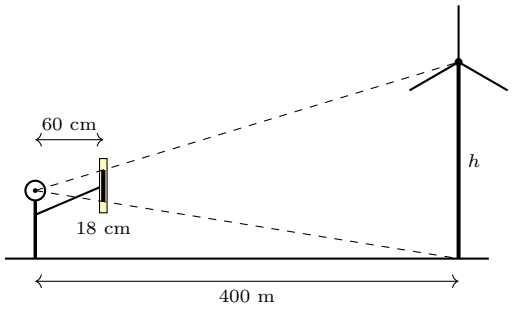
A ruler at arm's length. Hold a ruler straight up at arm's length. Close one eye. Slide the ruler so its 0 lines up with the bottom of the object, and read the ruler where the top of the object lines up. The two sight lines from your eye make two triangles, one inside the other. The ruler and the object are both upright, so they are parallel, and the triangles are similar (as in the nested triangles of Lesson 5).

The ruler method. $\dfrac{\text{height of object}}{\text{distance to object}} = \dfrac{\text{ruler reading}}{\text{eye to ruler}}$

The ruler reading and the eye-to-ruler distance must be in the same unit (both cm). Then their ratio has no unit, and the height comes out in the unit of the distance.

Example 5

Paige is 400 m from a wind turbine in southern Alberta (she found the distance on a map app). She holds a ruler upright 60 cm from her eye. The tower, from the ground to the hub in the centre of the blades, covers 18 cm of the ruler. How tall is the tower?

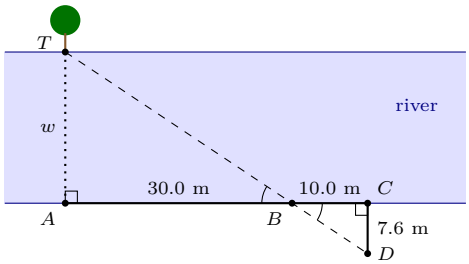


Stakes across a river. You can't stretch a tape across a river, but you can make a small copy of the triangle on your side. Pick a tree T on the far bank. Put stake A straight across from it. Walk along the bank to stake B, then on to C. At C, turn 90° and walk away from the river until B lines up with the tree: that is point D. The triangles TAB and DCB each have a right angle, and the angles at B are **vertically opposite** (Unit 5), so they are equal. The triangles are similar.

Example 6

Kenji uses the stake method to find the width w of the Bow River at a fishing spot. He measures $AB = 30.0$ m, $BC = 10.0$ m and $CD = 7.6$ m.

(a) Name the matching sides: TA matches _____, and AB matches _____.



(b) How wide is the river?

4. Is the Answer Reasonable?

Indirect measurement is easy to get wrong, and nobody can climb up to check. So always check your answer.

Checks for indirect measurement

- **Compare inside one triangle.** If the stick's shadow is longer than the stick, then *every* shadow is longer than its object at that moment. A tree with a 21 m shadow must then be shorter than 21 m.
- **Estimate with the scale factor.** If one shadow (or distance) is about 30 times the other, the heights differ by about 30 times.
- **Compare with something you know.** One storey of a building is about 3 m. A school flagpole is often 8 to 12 m. A tree that is 90 m tall is not reasonable in Alberta.

Your answer can also be a little off because of the measurements: shadows measured at different times, ground that is not level, an object that leans, a mirror that is not flat, or using your full height instead of your eye height. Small mistakes in short lengths (the stick, your eye height) grow when you multiply up.

Example 7

Each student made a mistake. Explain the mistake, and give the correct answer if you can.

(a) Ava: a 2.0 m post has a 3.0 m shadow. A tree's shadow is 21 m. She writes $\frac{h}{21} = \frac{3.0}{2.0}$, so $h = 31.5$ m.

(b) Ethan is 1.80 m tall and his eyes are 1.68 m high. Mirror method: he is 2.1 m from the mirror, and the mirror is 14.7 m from a tree. He writes $h = 1.80 \times 14.7 \div 2.1 = 12.6$ m.

(c) Mia measured her shadow at 9:00 a.m. and the school's shadow at 2:00 p.m. She used them in the shadow method.

Lesson 7: Unit 6 Review

Every rule from the unit on one page, worked examples that mix the topics, and a practice unit test.

1. The Unit on One Page

Use this page to study. Each row is one lesson: what to do, and a short example.

Skill	What to do	Example
Similar polygons and scale factor	**Similar** figures have the same shape: matching (**corresponding**) angles are equal and matching sides are in the same ratio. Name them in matching order: $ABCD \sim PQRS$ means $A \leftrightarrow P$, $B \leftrightarrow Q$, ... **Scale factor** k = new length $\div$ original length. New = $k\times$ original; original = new $\div k$. $k > 1$: enlargement; $k < 1$: reduction; $k = 1$: same size (congruent).	A 4 cm side becomes 10 cm: $k = 10 \div 4 = 2.5$ (enlargement) A 6 cm side becomes $2.5 \times 6 = 15$ cm
Are they similar?	Check **both**: (1) every pair of corresponding angles is equal, and (2) every side ratio (new $\div$ original) is the same number. One different ratio or one different angle means **not similar**. Regular polygons with the same number of sides are always similar.	3×5 and 6×9 rectangles: $6 \div 3 = 2$, $9 \div 5 = 1.8$ Ratios differ: not similar
Drawing enlargements and reductions	Multiply **every** length by k and keep **every** angle the same. Grid method: multiply each side (in squares) by k. Projection method: from a point O, $OA' = k \times OA$ along each ray.	$k = 1.5$: a 4-square side becomes $1.5 \times 4 = 6$ squares; a 90° corner stays 90°
Scale diagrams: maps and floor plans	A scale can be written in words (1 cm represents 5 m), as a ratio with the **same units** on both sides (1 : 500), or as a bar scale. Real = diagram $\times$ scale number; diagram = real $\div$ scale number. Change to the same unit first (1 m = 100 cm, 1 km = 100 000 cm).	Plan at 1 : 500, a wall 3.2 cm long: $3.2 \times 500 = 1600$ cm = 16 m
Similar triangles	Two triangles are similar if **two pairs of angles** are equal (AA): the third angles match because each triangle adds to 180°. Right triangles that share an acute angle are similar. A small triangle cut off by a line parallel to one side is similar to the big one. Then (side in big) $\div$ (matching side in small) is the same number k for every pair.	Small: 3 cm and 5 cm; big: 4.5 cm and x $k = 4.5 \div 3 = 1.5$ $x = 1.5 \times 5 = 7.5$ cm
Indirect measurement	**Shadows**: at the same time of day, height $\div$ shadow is the same for every object. **Mirror**: the angle in equals the angle out, so eye height $\div$ (you to mirror) = object height $\div$ (mirror to object).	1.6 m person, 2 m shadow; a tree's shadow is 15 m: $h = 1.6 \div 2 \times 15 = 12$ m

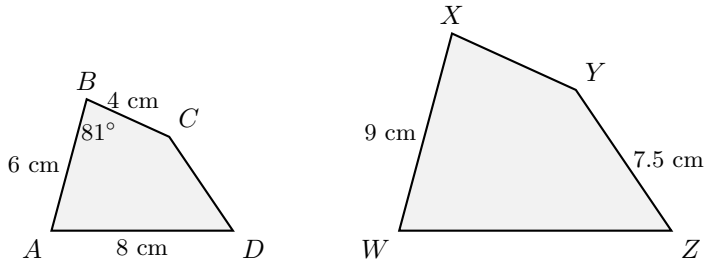
Remember. Match the sides in the **same order** as the names ($AB \leftrightarrow PQ$). Put the same figure on top in every ratio (big over small, or small over big). Lengths change by k; angles **never** change. Change to the same unit before you use a scale. Check: an enlargement must give bigger lengths, a reduction smaller ones.

Bonus fact. When the lengths are multiplied by k, the **area** is multiplied by $k \times k$. Double every side of a 3 m by 4 m patio and you get 6 m by 8 m: $48 \div 12 = 4$ times the area, not 2 times.

2. Similar Polygons and Scale Factor

Example 1

$ABCD \sim WXYZ$. (The diagrams are not to scale.)



(a) Write the side of $WXYZ$ that matches each side of $ABCD$, and the angle that matches $\angle B$.

$AB \leftrightarrow$ _____ $BC \leftrightarrow$ _____
$CD \leftrightarrow$ _____ $DA \leftrightarrow$ _____
$\angle B \leftrightarrow$ _____

(b) Find the scale factor from $ABCD$ to $WXYZ$. Is it an enlargement or a reduction?

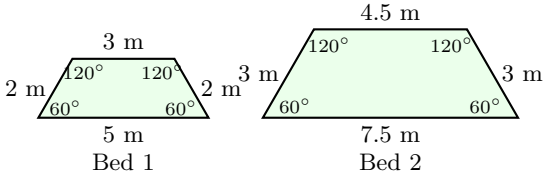
(c) Find XY, ZW and $\angle X$.

(d) Find CD.

3. Are They Similar?

Example 2

(a) Two flower beds are shaped like trapezoids. Are they similar? Show both tests.



Sides (Bed 2 ÷ Bed 1)	Ratio
bottom: $7.5 \div 5$	
top: $4.5 \div 3$	
left: $3 \div 2$	
right: $3 \div 2$	

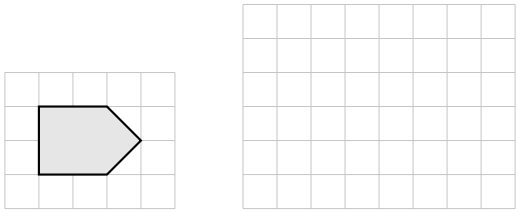
(b) Bree orders a 10 in by 12 in print of a 6 in by 9 in photo. Are the two rectangles similar? What will happen to her picture?

(c) A square and a rhombus both have 5 cm sides. The rhombus has angles of 60° and 120°. Are they similar? Explain.

4. Drawing Enlargements and Reductions

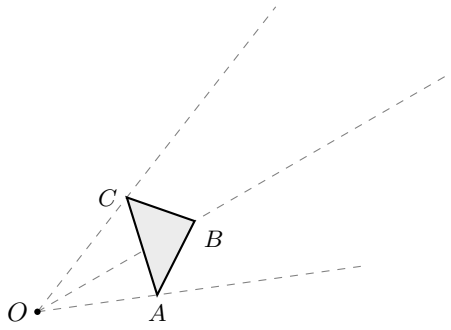
Example 3

(a) Enlarge the sign on the left with a scale factor of 2. Draw it on the grid on the right.



original $k = 2$

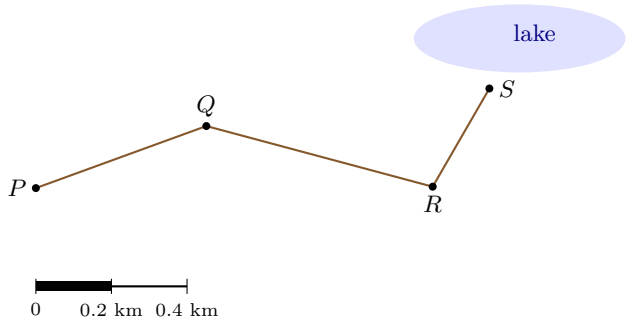
(b) Use the projection method from point O with $k = 2.5$. Find OA', OB' and OC', then draw $\triangle A'B'C'$. Here $OA = 2$ cm, $OB = 3$ cm and $OC = 2.4$ cm.



5. Scale Diagrams

Example 4

Here is part of a trail map in Kananaskis. The route goes from the trailhead P to Q, then R, then the lake shore at S. This map is printed at its real size, so you can measure it. Print this booklet at 100 % (actual size, not "fit to page") so your measurements match the answers.



(a) Write the scale in words and as a ratio.

(b) Measure the route. $PQ = 2.4$ cm, $QR = 3.1$ cm and $RS = 1.5$ cm. How far is the hike from P to S?

(c) A sign says the lake is 0.56 km long. How long should the lake be on the map? Measure to check.

Example 5

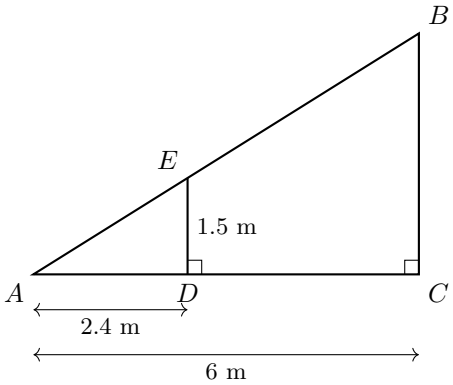
(a) A bedroom is 3.8 m by 4.4 m. Find its size on a floor plan with a scale of 1 : 50.

(b) A garage is 12 m by 9 m. The space for the drawing is 20 cm by 15 cm. Which scale should you use: 1 : 50, 1 : 100 or 1 : 200? Explain.

6. Similar Triangles

Example 6

(a) In $\triangle KLM$, $\angle K = 47°$ and $\angle L = 68°$. In $\triangle PQR$, $\angle P = 65°$ and $\angle R = 47°$. Are the triangles similar? If they are, name them in matching order.



(b) The side of a shed roof is a right triangle. A post DE stands straight up, so it is parallel to the end wall BC. Explain why $\triangle ADE \sim \triangle ACB$.

(c) Find the height of the wall, BC.

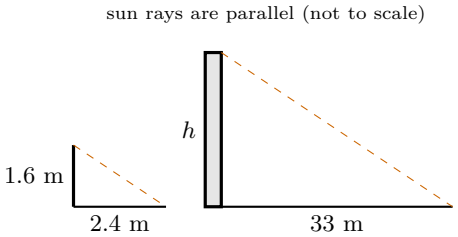
(d) Another post will stand 4.2 m from A. How tall must it be, to the nearest tenth of a metre?

7. Indirect Measurement: Shadows and Mirrors

Calculator. To find $1.6 \div 2.4 \times 33$, press 1.6 [÷] 2.4 [×] 33 [ENTER]. Keep the full value until the last step.

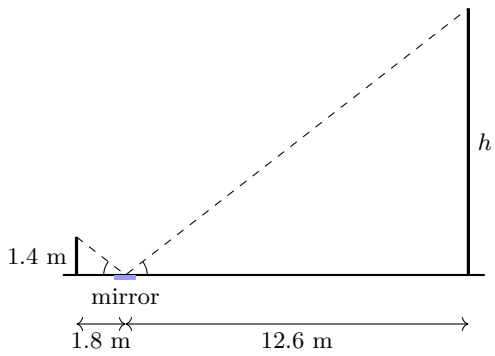
Example 7

At 3:00 in the afternoon, Priya is 1.6 m tall and casts a shadow 2.4 m long. At the same time, a grain elevator casts a shadow 33 m long. How tall is the elevator?



Example 8

Jordan wants the height of a light pole at the football field. He puts a mirror on the ground 12.6 m from the pole and steps back until he sees the top of the pole in the mirror. He is then 1.8 m from the mirror, and his eyes are 1.4 m above the ground.



(a) Why are the two triangles similar?

(b) Find the height of the light pole.